



MIDHUNESH.R

☎ +91 9789625329

✉ midhunesh6@gmail.com

🌐 linkedin.com/in/midhunesh-r

📍 114/1, Gandhinagar - 1,
Thekkupalayam(post)
Coimbatore - 641 020
Tamilnadu, India

TECHNICAL SKILLS

- C Programming
- Embedded C
- Microcontrollers
(ESP Arduino , Raspberry pi)
- Sensor Interfacing
- Basic Analog and Digital Electronics
- Microsoft Excel

AREA OF INTEREST

- Embedded Systems
- Microcontrollers & Sensor Interfacing
- Automotive Electronics
- Analog and Digital Electronics
- Internet Of Things (IoT)

SOFT SKILLS

- Teamwork
- Communication
- Problem-Solving
- Time Management
- Adaptability

LANGUAGES

- Tamil(native)
- English
- Hindi (basic)

CAREER OBJECTIVE

A tech-driven Electronics and Communication Engineering student with a strong foundation in electronics, embedded systems, microcontrollers, and digital technologies. Possessing knowledge of Embedded C programming, basic circuit design, and communication protocols such as I2C, SPI, and UART, along with a strong passion for hardware development, embedded systems, and electronic product design. Seeking an opportunity to apply technical skills in a practical engineering environment, gain hands-on experience in embedded system development, and contribute effectively to innovative electronic solutions while continuously enhancing professional and technical expertise.

EDUCATION

Bachelor of Engineering (B.E.) – Electronics and Communication Engineering

Dr. NGP Institute of Technology, Coimbatore **2023 – 2027**

Currently pursuing a Bachelor's degree in Electronics and Communication Engineering. Building knowledge in embedded systems, electronics, communication technologies, and programming through academic learning and projects.

Higher Secondary Certificate (HSC)

Sri Ramakrishna Mission Vidyalyaya , Coimbatore **2021 – 2023**

Completed Higher Secondary Education with a strong foundation in Mathematics and Science. Developed analytical and problem-solving skills that support my engineering studies.

CERTIFICATIONS & WORKSHOPS

50 Excel Champs – Excel Champs

DEC – 2024

Completed training on Microsoft Excel covering essential functions, data organization, and spreadsheet tools used for basic data analysis and reporting.

Analog Electronics – Udemy

SEP – 2024

Completed an online course covering the fundamentals of analog electronics including semiconductor devices, diodes, transistors, amplifiers, and basic electronic circuit concepts.

NextGen AIML in IoT Workshop – IIT Palakkad

FEB – 2024

Participated in a two-day workshop on AI/ML applications in IoT, gaining insights into smart systems, automation, and intelligent IoT solutions.

PROJECTS

STEM PROJECT – FIRE FIGHTING ROBOT

Designed and developed an autonomous fire-fighting robot using an Arduino microcontroller and multiple sensors. The system uses flame and infrared sensors to detect fire and locate its position, while ultrasonic sensors help in obstacle detection and safe navigation. The robot uses DC motors and motor drivers for movement and activates a water spraying mechanism to extinguish fire automatically. Arduino programming is used for sensor processing, navigation, and fire suppression control.

MINI PROJECT 1 : SMART CHECK-IN: Enhancing College Bus Attendance with RFID

Developed an RFID-based smart attendance system to automate and monitor student bus attendance efficiently. The system uses RFID cards and a microcontroller to identify students and record their attendance when they enter the bus. The attendance details are displayed on an LCD and stored digitally for monitoring and record keeping. The project helps reduce manual errors, improves accuracy, and enables faster attendance tracking during bus boarding. It demonstrates the practical application of embedded systems, sensor interfacing, and real-time data processing in transportation management.

MINI PROJECT 2 : SMART TALK – Gesture Based Communication System (ESP32)

Developed a gesture-based assistive communication device using an ESP32 microcontroller and APDS-9960 gesture sensor to help physically challenged individuals convey basic needs. The system detects hand gestures and converts them into messages such as Hungry, Thirsty, and Help, which are displayed on an LCD and transmitted to a caretaker via Bluetooth using the Serial Bluetooth Terminal mobile app. The device provides multimodal alerts through LED indicators, buzzer notifications, and voice output using a DFPlayer module and mini speaker. A buck converter with Li-ion battery supply ensures stable power for continuous operation.

INTERNSHIP

Enthu Technology Solutions India Pvt Ltd

Duration: 07/06/2025 – 22/06/2025

Gained practical exposure to embedded systems and IoT concepts using Raspberry Pi. Worked with GPIO interfacing, sensors, and Python scripts to acquire and process real-time data. Developed mini-projects focused on real-time monitoring and device control, enhancing understanding of hardware-software integration, sensor interfacing, and basic embedded programming. Participated in implementing and testing IoT-based applications, gaining practical knowledge of data acquisition and device communication. Assisted in troubleshooting hardware and software issues, improving problem-solving and debugging skills. The internship provided valuable hands-on experience in embedded development and strengthened understanding of real-world IoT solutions and system design.

DECLARATION

I hereby declare that the information provided above is true and correct to the best of my knowledge and belief. I assure that the information provided is accurate and complete.